

Are Code Reviews a Thing of the Past?

The backbone of a successful app or software is its impeccable code. This has to be kept pristine, else the app will fail. For code to be flawless, it has to be reviewed constantly.

This can be achieved manually or through the use of a dedicated tool. Some pretty decent tools in this area are FindBugs, PMD, CheckStyle and SonarQube.



The reason why legacy software survived the dynamics of maintenance for years was due to the discipline and diligence followed by the engineers. This is more of a culture, which was embedded in the system and eventually helped in maintaining the rhythm of the IT growth we have seen so far in India.

In the early stages of this revolution, software companies put together a lot of processes and tried their best to institutionalise these across the organisation. They took industry standards as benchmarks, and worked hard in getting the required certifications and assessments to label themselves as professional practitioners in software development and maintenance. Certifications like ISO, CMM, CMMi, etc, helped organisations in clinching some of those multi-million dollar outsourcing deals from MNCs by instilling the necessary trust and confidence.

Just taking this a few levels down, the software development life cycle included a lot of checks and controls to ensure that the quality was taken care of. This was made possible through the identification of bugs, both during compilation and at the run time. The analysis of source code, also called static analysis, helps in identifying the bugs prior to getting it to runtime through the testing team. There are some industry metrics that have evolved over a period of time, and these need to be evaluated on a continuous basis to stay relevant. Of course, they may depend on so many other

parameters and may depend on the project at hand too.

Just as we have proof-reading for all the writing exercises, there is a need to have something similar for the code we develop, and it has to be an independent exercise performed by somebody other than the actual developer. Overall, the intention is to catch all the blind spots that the developer might have overlooked. There may be many reasons for these issues to creep in, like the lack of enough experience, or the 'silo approach' of focusing on the immediate problem and ignoring the holistic perspective, etc. Though the experience may help in minimising some of these errors, it can be a challenge to reduce it by 100 per cent; so this exercise is important irrespective of the developer's experience.

Some of the aspects that can be handled as part of the code reviews include:

- Coding standards violation
- Inefficient use of programming constructs
- Possible code complexity
- Documentation

Though there are a few good tools to handle code reviews, these were purely manual when software development started out. Today, companies carry out a blend of both manual and tool-driven reviews to ensure broader coverage.

Typically, every experienced software development company has its own process established at the organisational level, which is followed by all the project teams. There is a separate department called SQA or Software Quality Assurance (the names may differ across different organisations), which oversees and facilitates the defining of standards and their implementation across the organisation.

There are enough guidelines and checklists defined at the organisation level to help project teams prepare for the code reviews. The same are used by the reviewers to evaluate the code quality. Also, SQA teams continuously work to identify the biggest trends in the industry, taking consent from the management; this is an ongoing process to keep them up to date with the latest changes. They also conduct regular education sessions to build awareness about these initiatives across the broader project teams on a regular basis.

As the complexity of the projects is increasing with the time to market coming down, it has become imperative to look out for alternatives to improve the quality rather than rely on the manual process. This has triggered the evolution of tools that can ensure broader coverage, improve time to market and, at the same time, align with the existing IDEs.